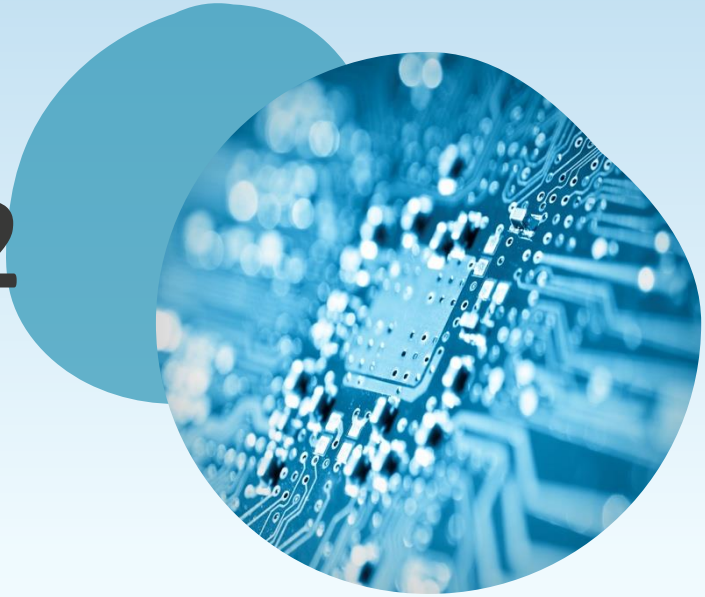
Interfacing MSP432 BOOSTERPACK SENSORS - I

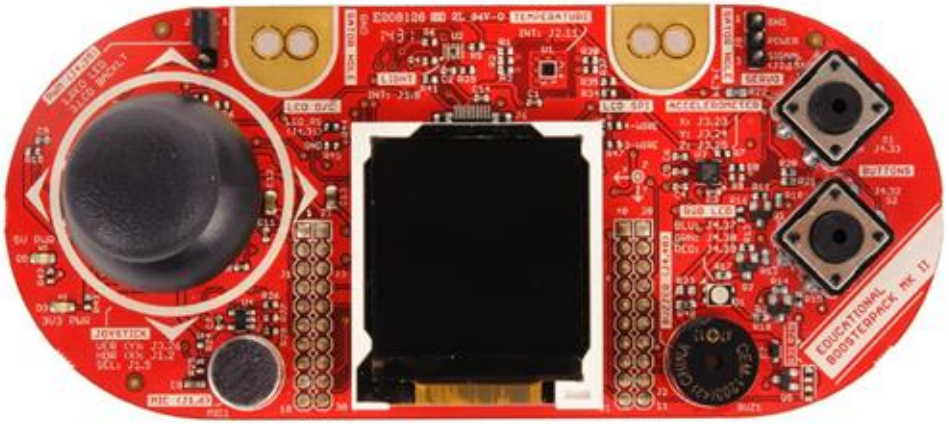


Embedded Software for the Internet Of Things

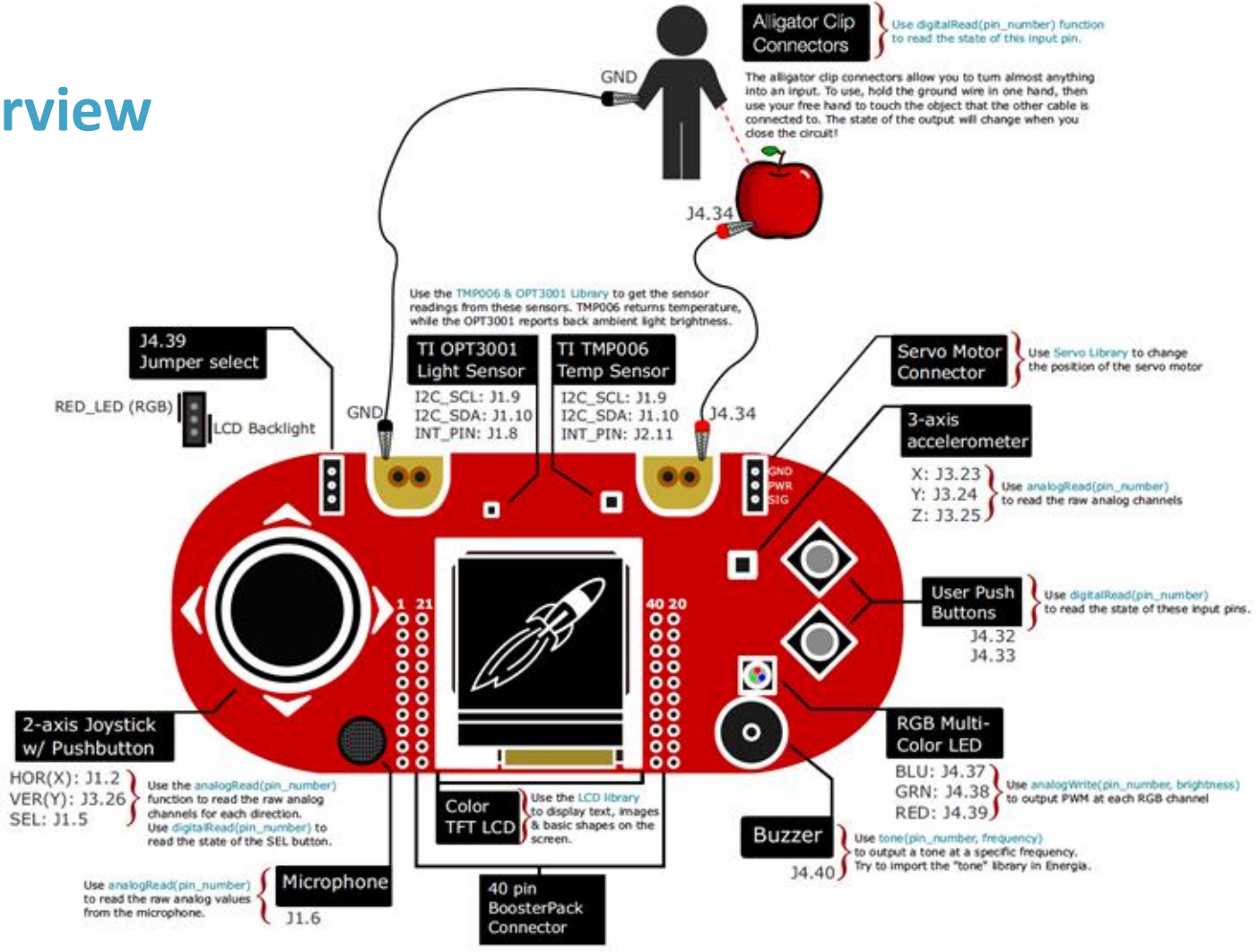
Kasim Sinan Yildirim

BOOSTXL-EDUMKII EDUCATIONAL BOOSTERPACK

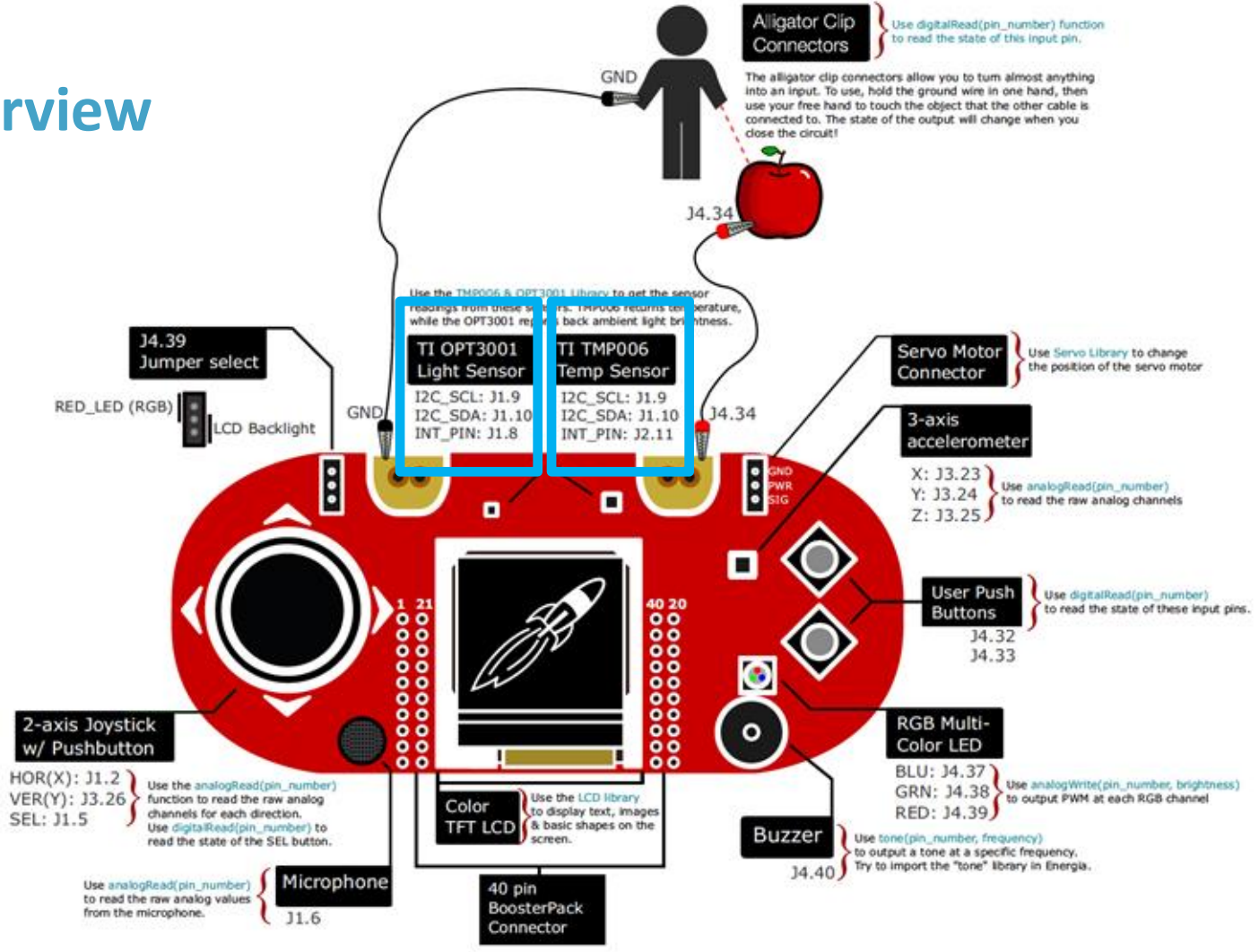
- Various analog and digital inputs and outputs
- TI OPT3001 Light Sensor
- TI TMP006 Temperature Sensor
- Servo Motor Connector
- 3-Axis Accelerometer
- User Push Buttons
- RGB Multi-color LED
- Buzzer
- Color TFT LCD Display
- Microphone
- 2-Axis Joystick with Pushbutton



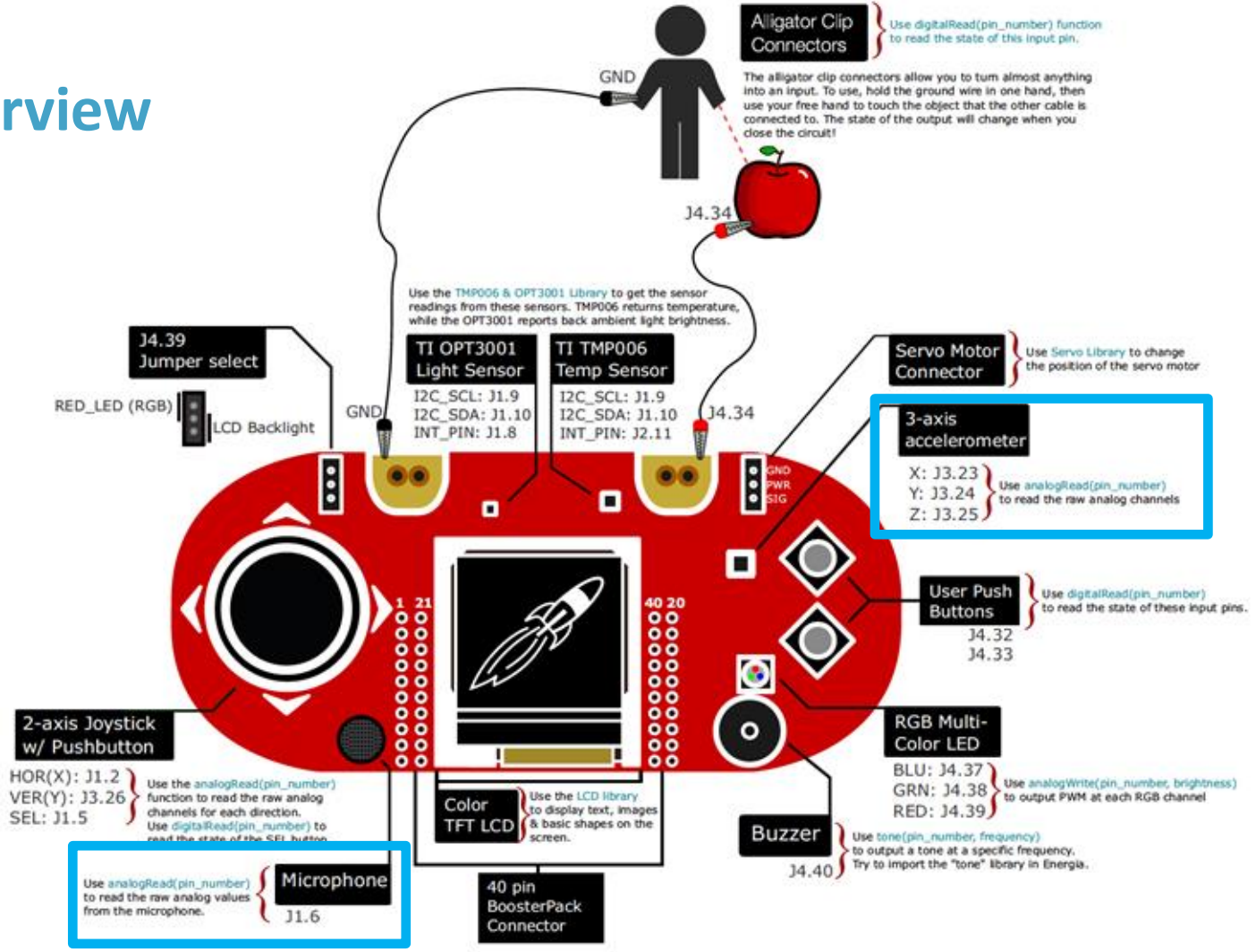
Sensor Overview



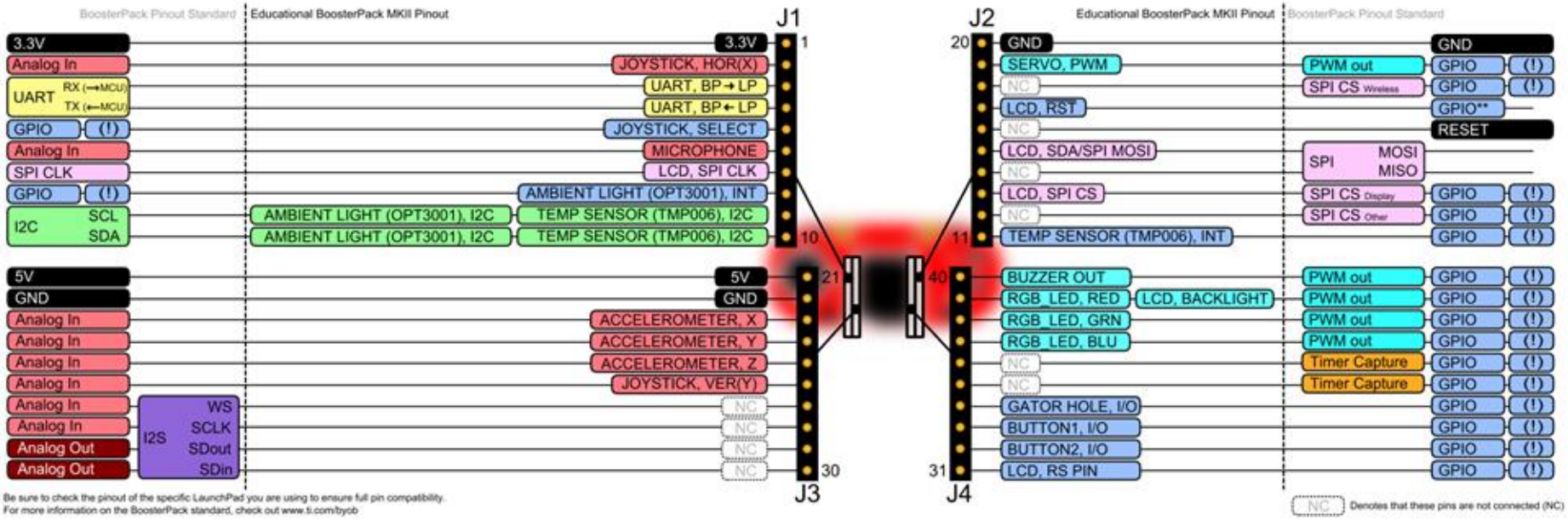
Sensor Overview



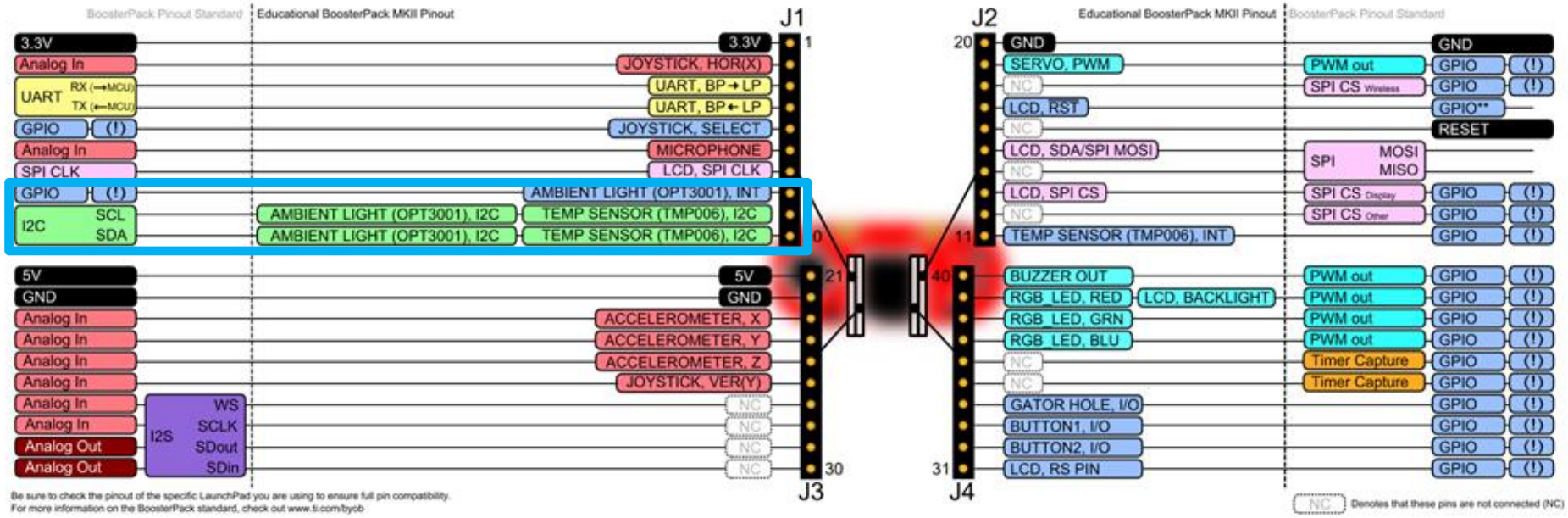
Sensor Overview



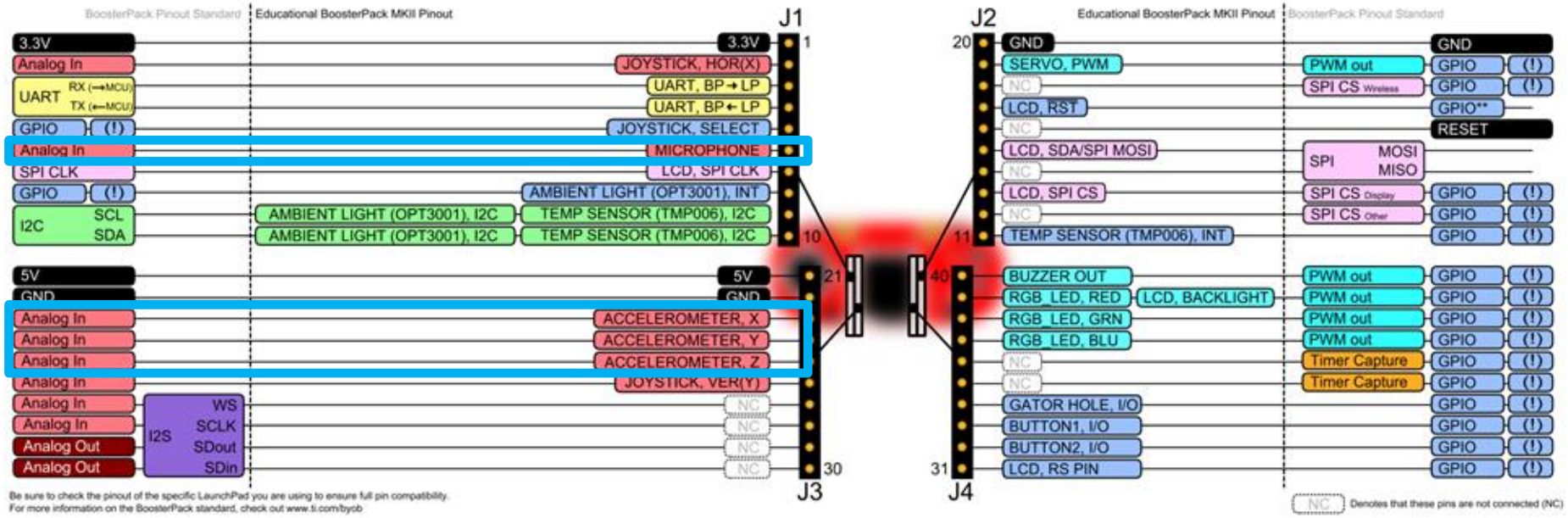
PinOut Diagram



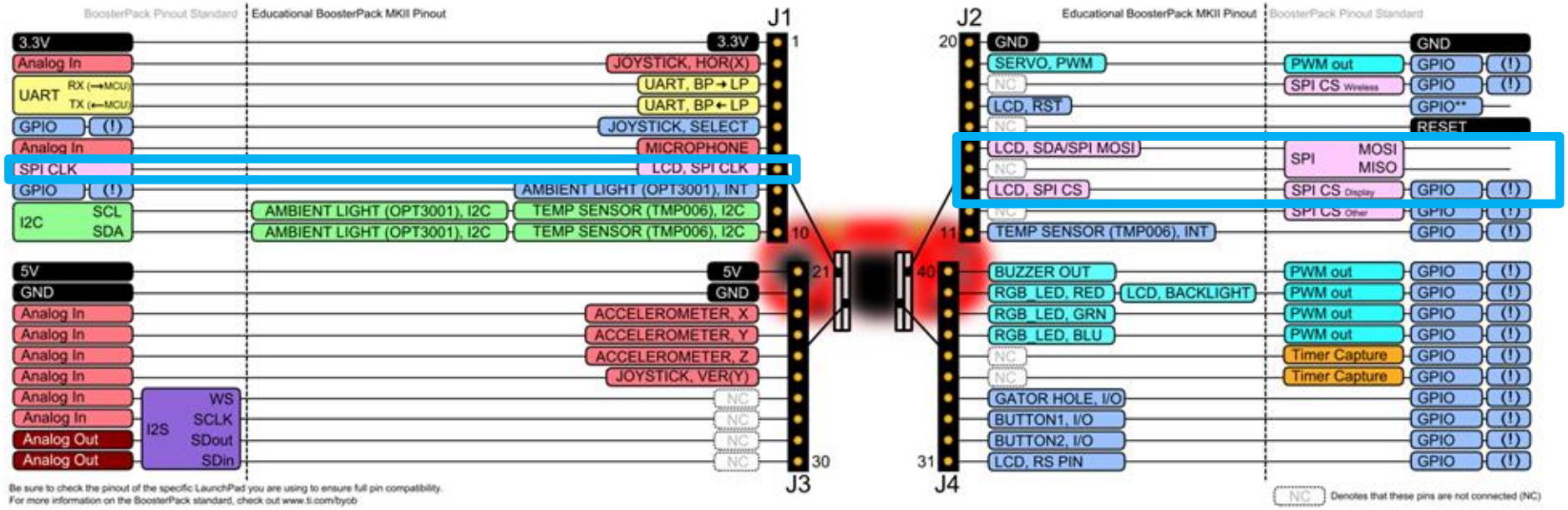
PinOut Diagram



PinOut Diagram


BoosterPack Pinout Standard
Educational BoosterPack MKII Pinout
J1
J2
Educational BoosterPack MKII Pinout
BoosterPack Pinout Standard
3.3V
Analog In
UART
RX (⟶MCU)
TX (⟵MCU)
GPIO
(!)
Analog In
SPI CLK
GPIO
(!)
I2C
SCL
SDA
5V
GND
Analog In
Analog In
Analog In
Analog In
Analog In
Analog Out
Analog Out
I2S
WS
SCLK
SDout
SDin
3.3V
JOYSTICK, HOR(X)
UART, BP→LP
UART, BP←LP
JOYSTICK, SELECT
MICROPHONE
LCD, SPI CLK
AMBIENT LIGHT (OPT3001), INT
AMBIENT LIGHT (OPT3001), I2C
TEMP SENSOR (TMP006), I2C
AMBIENT LIGHT (OPT3001), I2C
TEMP SENSOR (TMP006), I2C
5V
GND
ACCELEROMETER, X
ACCELEROMETER, Y
ACCELEROMETER, Z
JOYSTICK, VER(Y)
NC
NC
NC
NC
1
10
21
30
J3
20
11
40
31
J4
GND
SERVO, PWM
NC
LCD, RST
NC
LCD, SDA/SPI MOSI
NC
LCD, SPI CS
NC
TEMP SENSOR (TMP006), INT
BUZZER OUT
RGB_LED, RED
RGB_LED, GRN
RGB_LED, BLU
NC
NC
GATOR HOLE, I/O
BUTTON1, I/O
BUTTON2, I/O
LCD, RS PIN
LCD, BACKLIGHT
PWM out
SPI CS Wireless
GPIO
GPIO
GPIO**
RESET
SPI
MOSI
MISO
SPI CS Display
SPI CS Other
GPIO
GPIO
GPIO
PWM out
PWM out
PWM out
PWM out
Timer Capture
Timer Capture
GPIO
GPIO
GPIO

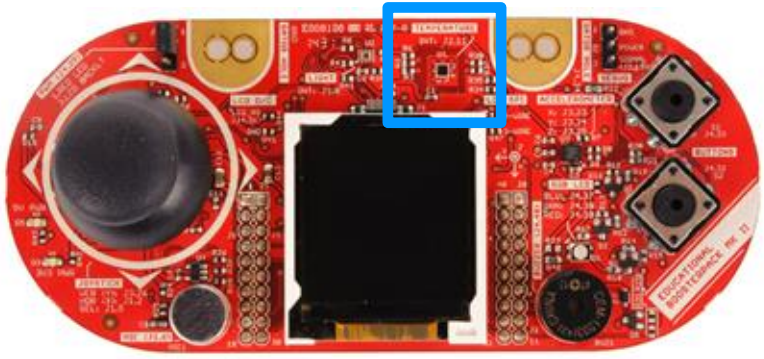
PinOut Diagram


BoosterPack Pinout Standard
Educational BoosterPack MKII Pinout
J1
J2
Educational BoosterPack MKII Pinout
BoosterPack Pinout Standard
3.3V
Analog In
UART
RX (⟶MCU)
TX (⟵MCU)
GPIO
(!)
Analog In
SPI CLK
GPIO
(!)
I2C
SCL
SDA
5V
GND
Analog In
Analog In
Analog In
Analog In
Analog In
Analog In
Analog Out
Analog Out
I2S
WS
SCLK
SDout
SDin
3.3V
JOYSTICK, HOR(X)
UART, BP→LP
UART, BP←LP
JOYSTICK, SELECT
MICROPHONE
LCD, SPI CLK
AMBIENT LIGHT (OPT3001), INT
AMBIENT LIGHT (OPT3001), I2C
TEMP SENSOR (TMP006), I2C
AMBIENT LIGHT (OPT3001), I2C
TEMP SENSOR (TMP006), I2C
5V
GND
ACCELEROMETER, X
ACCELEROMETER, Y
ACCELEROMETER, Z
JOYSTICK, VER(Y)
NC
NC
NC
NC
1
10
21
30
J3
20
11
40
31
J4
GND
SERVO, PWM
NC
LCD, RST
NC
LCD, SDA/SPI MOSI
NC
LCD, SPI CS
NC
TEMP SENSOR (TMP006), INT
BUZZER OUT
RGB_LED, RED
RGB_LED, GRN
RGB_LED, BLU
NC
NC
GATOR HOLE, I/O
BUTTON1, I/O
BUTTON2, I/O
LCD, RS PIN
LCD, BACKLIGHT
PWM out
SPI CS Wireless
GPIO**
RESET
SPI
MOSI
MISO
SPI CS Display
SPI CS Other
PWM out
PWM out
PWM out
PWM out
Timer Capture
Timer Capture
GND
GPIO
(!)
GPIO
GPIO
GPIO

TMP006 INFRARED CONTACTLESS TEMPERATURE SENSOR

- Measures the temperature without the need to make contact with the object.
  - Uses a thermopile to absorb the **infrared energy** emitted from the object
    - A thermopile is an electronic device that converts thermal energy into electrical energy.
    - The change in thermopile voltage to determines the object temperature.

BoosterPack Plug-in Module Header Connection	Pin Function
J1.9[(1)]	I²C SCL
J1.10[(1)]	I²C SDA
J2.11	TMP006 Interrupt

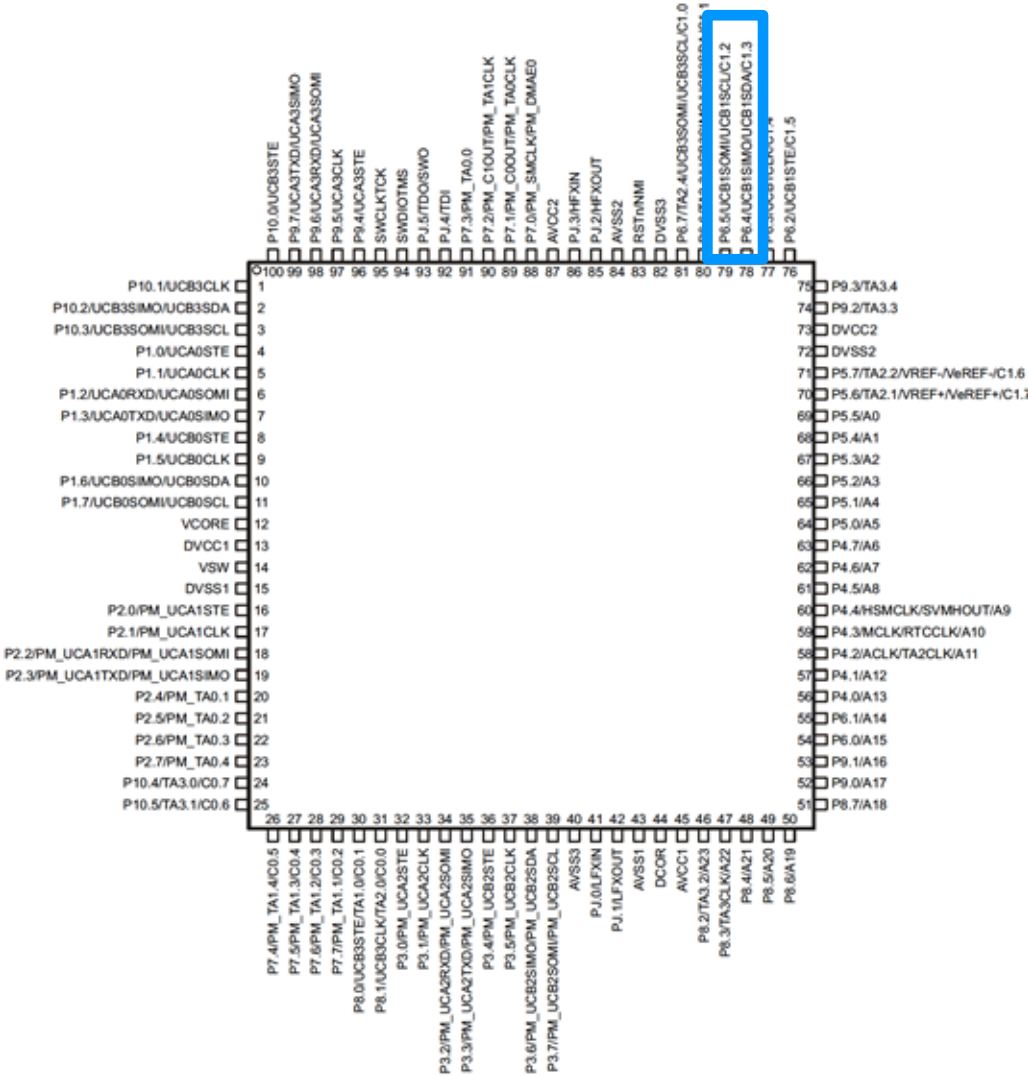


Pin Mappings

- Launchpad pins:
  - P6.5: I2C_SCL
  - P6.4: I2C_SDA



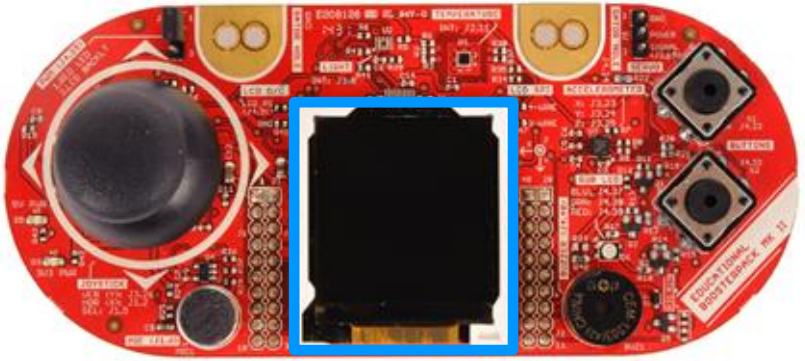
PIN NAME (P6.x)	x	FUNCTION	CONTROL BITS OR SIGNALS[1]		
			P6DIR.x	P6SEL1.x	P6SEL0.x
P6.2/UCB1STE/C1.5[2]	2	P6.2 (I/O)	I: 0; O: 1	0	0
		UCB1STE	X[3]	0	1
		N/A	0	1	0
		DVSS	1		
		C1.5[4][5]	X	1	1
P6.3/UCB1CLK/C1.4[2]	3	P6.3 (I/O)	I: 0; O: 1	0	0
		UCB1CLK	X[3]	0	1
		N/A	0	1	0
		DVSS	1		
		C1.4[4][5]	X	1	1
P6.4/UCB1SIMO/UCB1SDA/C1.3[2]	4	P6.4 (I/O)	I: 0; O: 1	0	0
		UCB1SIMO/UCB1SDA	X[3]	0	1
		N/A	0	1	0
		DVSS	1		
		C1.3[4][5]	X	1	1
P6.5/UCB1SOMI/UCB1SCL/C1.2[2]	5	P6.5 (I/O)	I: 0; O: 1	0	0
		UCB1SOMI/UCB1SCL	X[3]	0	1
		N/A	0	1	0
		DVSS	1		
		C1.2[4][5]	X	1	1



LCD Display

- 128x128-pixel TFT LCD
  - supports display updates up to 20 frames per second (FPS)
  - controlled through the SPI interface
  - has a color depth of 262K colors and a contrast ratio of 350

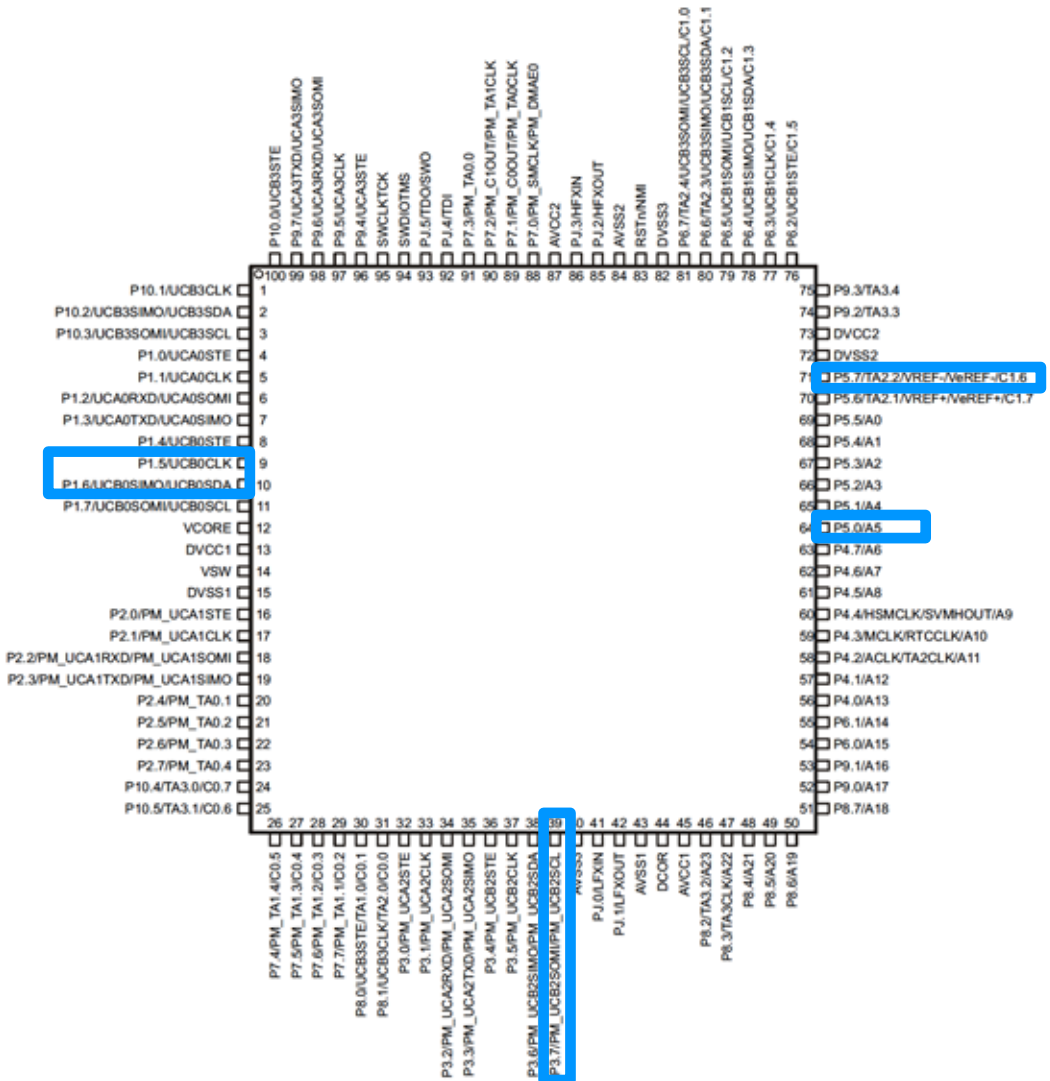
BoosterPack Plug-in Module Header Connection	Pin Function
J1.7	LCD SPI clock
J2.13	LCD SPI chip select
J2.15	LCD SPI MOSI
J4.31	LCD reset pin
J4.39[(1)]	LCD backlight



Pin Mapping

- SCK_PORT: P1.5
- MOSI_PORT: P1.6
- RST_PORT: P5.7
- CS_PORT: P5.0
- DC_PORT: P3.7





Lab #1

- Perform temperature sensing via timers
  - Set a 1 second timer and update the LCD accordingly.
  - Display the temperature in Celsius.